Imperial College London

Department of Bioengineering

Assessment title: Thesis

Simulation-Guided Design and Kinetic Studies for the Automation of Nanocatalyst Amplified Lateral Flow Device

By

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Master of Research (MRes) in Bioengineering

Words counts: 6064

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Date: September 2020

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Simulation-Guided Design and Kinetic Studies for the Automation of Nanocatalyst Amplified Lateral Flow Device

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KEWORDS: lateral flow immunoassay, nanocatalyst amplification, device automation, one-dimensional flow rate modeling, kinetic modeling, point-of-care, infectious disease detection

ABSTRACT: The development of point-of-care (PoC) diagnostics is the key to detecting infectious disease, preventing disease transmission, and providing prompt treatment in resource-limited settings. Being one of the most commonly used PoC techniques, paper-based lateral flow immunoassay (LFIA) is yet limited to early disease diagnosis due to its insufficient sensitivity in short time frames. In light of this, catalytic nanomaterials have been exploited as absorbance-based LFIA labels for signal amplification and enhanced sensitivity. However, the application of such nanomaterials-based LFIA in PoC diagnostics still requires optimization as the process often involves specialized personnel to perform multiple steps of preparation and addition of reagents. To address this, this thesis presents the integration of paper-based microfluidics simulation and kinetic modeling with nanocatalyst-based LFIA, verifying the feasibility of achieving an automated LFIA device with complex sample processing and sensitive signal amplification. The synthesized platinum nanoparticles (PtNPs) exhibited efficient peroxidase-like activity in catalyzing the oxidation of chromogenic substrates in the presence of hydrogen peroxide. These nanocatalysts might be readily adapted and modified with recognition elements as highly amplifying labels in the LFIA platform. Additionally, a three-channel LFIA device was designed to allow a sequential flow of reagents at precise time frames based on their one-dimension flow rate simulation result. Their flow behavior was expressed in mathematical models as a function of flow velocity over distance and distance over time. For the kinetic studies, a simulation model that theoretically described the binding kinetic of conventional antibodies and nanobodies in response to time was established based on the developed LFIA. This may complement the theoretical foundation for the selection of recognition elements as well as the estimation of binding responses and target concentration in an automated LFIA platform. These simulation models can be improved and capitalized for the detection of other biomolecules to facilitate the development of sensitive screening tools for infectious disease diagnosis in PoC settings.

1. INTRODUCTION

Sensitive and accurate diagnostics is the first step for appropriately treating disease. The accessibility of such diagnosis for early detection, particularly at resource-limited or on-site settings, is critical to the effectiveness of transmission prevention and appropriate treatment.[1] The application of point-of-care (PoC) diagnostics in early disease diagnosis (typically with a low concentration of target analytes) is often constrained due to its insufficient sensitivity over short time frames. Enzymes have been extensively used in colorimetric biomarker detection in laboratory settings as enhancers of diagnostic performance. However, their catalytic ability is highly susceptible to storage and operational condition.[2] The application of enzyme-based detection in PoC settings is limited. To address this, scholars have focused on the development of nanotechnology; alternative synthetic materials that possess intrinsic enzyme-like activity.[3,4] Having both the advantages of low-cost and high stability over a wide range of temperature and pH,[5] the development of these novel enzyme mimetics as amplification techniques for PoC diagnostics is an active field in academic research.

Recently, our group reported the synthesis of platinum nanocatalysts (PtNCs) that retain effective catalytic activity in serum samples. Thanks to their high catalytic efficiency and stability, they were employed as amplifying labels in paper-based lateral flow immunoassay (LFIA) to enhance its sensitivity. Such assay demonstrated outstanding sensitivity (in a femtomolar range) that surpassed the sensitivity of existing paper-based methods for the detection of p24, one of the earliest biomarkers of HIV.[6] Nonetheless, such nanomaterial-based diagnostic platform often requires specialized personnel to perform multiple steps, including a series of reagents preparation and addition. To overcome this complication, this thesis presents the simulation of paper-based microfluidics and kinetic modeling, so that complex sample processing, fluid handling, and sensitive signal amplification can be achieved autonomously. The significance of this thesis is to supplement nanocatalyst-amplified LFIAs, and hence to contribute to the development of a complete automated and sensitive biosensing platform from blood-in to results-out.

The objectives of this thesis were to synthesis platinum nanoparticle and investigate its catalytic activity (Appendix B); to simulate the sequential flow of nanocatalyst-amplified LFIA components and characterize the device design; to study the binding kinetics of the developed method based on kinetic modeling.

1.1. Infectious disease diagnosis

In infectious disease diagnosis, the health status of patients is examined by monitoring the biomolecular changes within their body. The detection of relevant biological markers with high sensitivity and specificity is critical to achieving early diagnosis of infectious diseases so as to effectively prevent global transmission and provide appropriate treatment.[7] Emerging infectious diseases (EIDs) have been regarded as a major disease burden on public health and economics worldwide.[8,9] EIDs are continuously emerging (e.g., COVID-19), resulting in outbreaks that threaten human health due to the limitations of existing diagnostic tools. For instance, most of diagnostic tools are often laboratory-based, requiring sophisticated infrastructure, advanced equipment, stable energy supply for storage and operation, and highly trained personnel, all of which are generally unavailable in remote areas and resource-limited settings.

1.1.1. Point-of-care diagnostics

In view of this, point-of-care (PoC) diagnostic platforms have the potential to make diagnostics accessible to all levels of the healthcare system in both developing and developed nations. They are regarded as cost-effective and rapid tools that do not require the use of laboratory infrastructure. In seconds to hours, these tests are performed in patients' homes or "near patients" in a hospital, clinic, or practitioner's office, involving minimal or no pre-preparation and technical training before use.[10] As such, PoC diagnostic platform can be applied at the primary care level for immediate clinical decision-making for disease control strategies, enhanced efficiency of the healthcare system, and improved patient outcomes. The WHO has created a set of criteria for PoC diagnostic test to address the need of infectious disease control, known as REASSURED ("Real-time connectivity, Ease of specimen collection, Affordable, Sensitive, Specific, User-friendly, Rapid and Robust, Equipment free or simple and Environmentally friendly, Deliverable to end-user").[11,12] Therefore, the design and development of devices that satisfy the REASSURED criteria are crucial to PoC diagnostics.

1.2. The use of LFIA as a PoC platform

One of the most prevalent techniques for rapid disease diagnosis particularly for PoC application, known for its low-cost and ease of use, is paper-based lateral flow immunoassay (LFIA). Based on movement driven by capillary force, LFIA employs a chromatographic system, immunochemical reaction, and usually a signal-generating system to produce a rapid visual readout.[13] Its standard structure is composed of a sample pad, a conjugate pad, a nitrocellulose membrane, and an absorbent pad.[14] The addition of a sample containing analytes onto the sample pad initiates the test *via* capillary action. The sample first migrates to the conjugate pad, where the primary biomolecule, dried and labeled with tags, is being rehydrated. The analyte interacts with the conjugate as they flow along the nitrocellulose membrane, which has been immobilized with a line of affinity reagents specifically designed to capture the labeled complex as the mixture migrates over. The secondary biomolecule modified on the control line can validate device functionality and the absorption pad absorbs excessive reaction mixture. With the labeling tags, users can evaluate the presence or absence of analytes by reading the signals (absorbance, fluorescence, and so on) generated from the test line (Figure 1).[15]

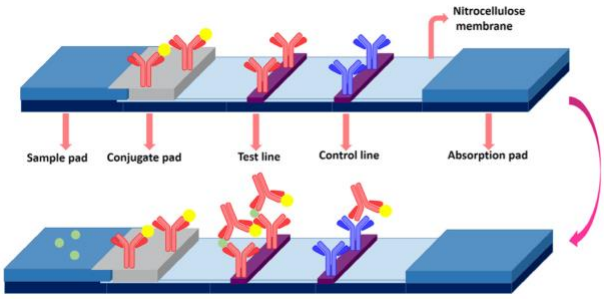

Figure 1. Schematic illustration of a standard sandwich lateral flow assay, where the primary and secondary antibodies serve as recognition elements. Figure reproduced from Bahadır and Sezgintürk.[15]

1.2.1. Nanomaterials-based colorimetric LFIA

Various nanomaterials have been exploited in immunoassays as signal generation tags to improve the sensitivity of the assays. The typical materials used in the past decades are the use of europium (III) nanoparticle,[16] nanocrystals with releasable fluorophores[17], quantum dots[18], and so on. Although these approaches can provide relatively higher sensitivity than visual labels, the fluorescent readout requires additional equipment for test result analysis, which is not practical for disease detection at PoC.

Nanomaterials-based colorimetric immunoassays are a promising alternative that does not require the use of additional equipment. Nanomaterials often serve as colorimetric labels and provide intense visual signal contrast due to localized surface plasmon resonance (LSPR). The general approaches include analyte-induced aggregation or morphology change of nanomaterials, nanomaterials consisting of loading enzyme labels, and nanomaterials with enzyme-like activity to catalyze chromogenic reactions.[19] Being the most commonly used labels, colloidal gold[20] and gold nanoparticles (AuNPs)[21] have been widely applied in LFIA because of their high LSPR, excellent physiochemical properties, and the simplicity of being functionalized with biomolecules using chemisorption and physisorption.[22]

1.2.2. Nanocatalyst with enzyme-like activity

To complement the above colorimetric LFIA, nanocatalyst have been increasingly applied as enzyme-mimetics. The pioneering work by Gao et al. demonstrated that ferromagnetic nanoparticles (MNPs) possess intrinsic peroxidase-like activity (Figure 2) and have higher thermal and pH stabilities over a wide range of conditions than natural enzyme such as horseradish peroxidase (HRP).[23] Integrated with chromogenic substrates, other nanomaterials with peroxidase-like activity such as modified AuNPs[24], platinum nanoparticles (PtNPs)[25], and Au@PtNPs[26] have also been utilized as signal amplification labels in LFIA to lower its detection limit.

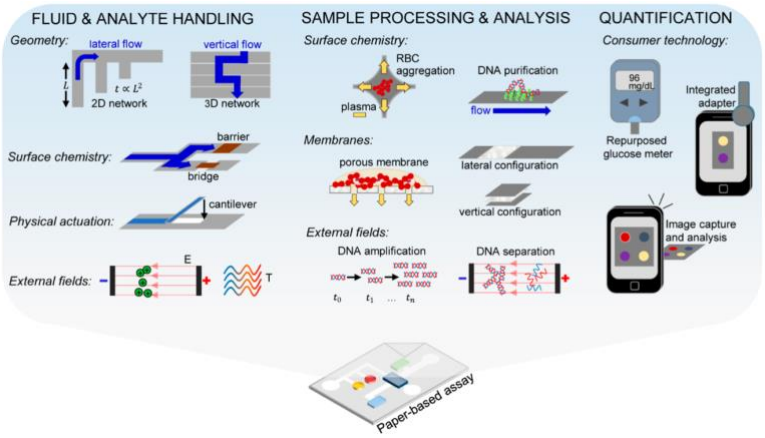

Figure 4. Diagram of the potential analytical capabilities (a) fluid and analyte handling (b) sample processing and analysis (c) result quantification of paper-based microfluidics for a wide range of applications in PoC detection. Figure adapted from Gong and Sinton.[29]

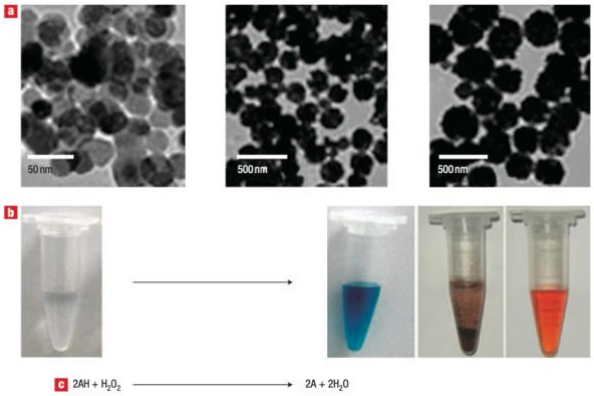

Figure 2. (a) Transmission electron microscopy (TEM) image of MNPs in different sizes (from left to right: 30, 150, and 300 nm). (b) The Fe_3O_4 MNPs demonstrate peroxidase-like activity toward various chromogenic substrates, including (from left to right) 3,3′,5,5′-tetramethylbenzidine (TMB), diaminobenzidine (DAB), and o-phenylenediamine (OPD) in the presence of H_2O_2. (c) Schematic illustration of the catalytic mechanism by Fe_3O_4 MNPs, in which AH represents the chromogenic substrate as hydron donor. Figure adapted from Gao et al.[23]

1.3. Amplified LFIA based on the peroxidase-like activity of PtNCs

Thanks to the high catalytic efficiency and stability of noble-metal nanoparticles as discussed above, they have been increasingly exploited in absorbance-based signal amplification systems for LFIA. Loynachan et al.[6] reported a serum-stable and nanoparticle labeled LFIA for the detection of HIV-1 p24 capsid protein, one of the earliest and remarkable biomarkers of HIV. By utilizing the peroxidase-like activity of porous platinum core-shell nanocatalysts (PtNCs), they achieved a 100-fold sensitivity improvement with a linear dynamic range in plasma and clinical samples (Figure 3). The antibody-functionalized PtNCs were incorporated with modified nanobodies in a sandwich format to allow target-dependent PtNCs' deposition at the test line. In the presence of H_2O_2, the deposited PtNC amplified the LFIA through catalyzing the oxidation of 4-chloro-1- naphthol/3,3'-diaminobenzidine, tetrahydrochloride (CN/DAB) to produce an insoluble black product for a positive sample. As a result, a colorimetric and quantitative HIV detection was achieved in a simple, robust, sensitive, and timely manner. However, further adaptation is required for PoC application as the experiment was performed in a dipstick format; multiple steps were involved in the manual preparation and addition of rinse and signal amplification reagents by well-trained personnel. To capitalize on the PtNCs amplified LFIA for use at PoC, fully automation of the assay is preferred.

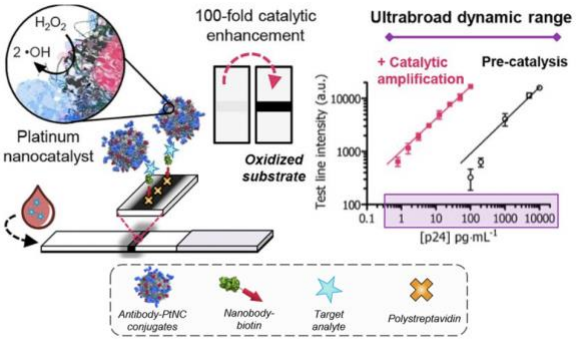

Figure 3. Schematic illustration of the PtNCs amplified LFIA. The interaction among antibody-PtNCs, target analytes, and biotinylated nanobody fragments in a sandwich format enables target deposition at the test line. The PtNCs deposited in the test line further catalyze the reaction between CN/DAB substrate and H2O2 to develop a black product for signal amplification. Figure adapted from Loynachan et al.[6]

1.4. Paper-based microfluidics

To achieve the automation of LFIA, microfluidic techniques are considered due to their advantages with complex sample processing, manipulation of liquids, precise reaction time control, and ease of use.[27] They have been incorporated with multiple techniques within a single device, including blood chemistry, nucleic acid analysis, and lateral flow immunoassay.[28] Particularly, paper-based microfluidics has significantly contributed to the miniaturization and automation of LFIA devices via the integration of sample preparation and operational steps into a self-integrated cartridge. Within such a cartridge, all reagents can be stored in advance, analysis can be

automated through precise flow control, and "hand-on" time and operational error are minimized. The development of the analytical ability of paper-based microfluidics over the past years was reviewed (Figure 4).[29]

One of the successful integrations between LFIA and paper-based microfluidic format is a two-dimensional paper network (2DPN). Yager and co-workers postulated that such integration could achieve sequential delivery of multiple reagents by enabling sophisticated fluid control.[30] In 2020, Byers et al. revealed an integrated 2DPN and LFIA device that leveraged complex solution inputs and utilized fully dried solutions of DAB, H_2O_2 and signal enhancement elements (Figure 5).[31] Notably, they applied mathematical models to optimize the flow control and to guide the device design. Based on their experiment and one-dimensional flow modeling result, they characterized their device dimensions to achieve a fully dried, automated detection platform with 116% signal enhancement for nucleic acid (NA) samples. However, the device design is not applicable for the production of lateral flow strips due to the complication of "leg design", so this thesis focuses on its merits of mathematical modeling on flow rate analysis.

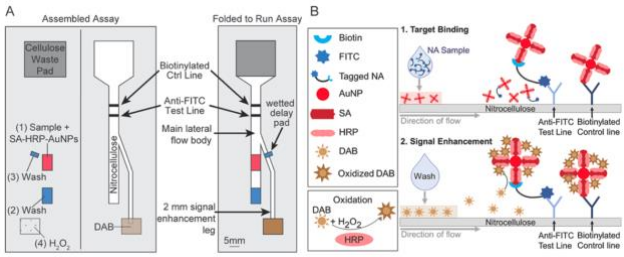

Figure 5. (A) Overview of an integrated 2DPN LFIA, showing the position of different reagents pads. (B) Schematic showing the strategies of target binding and signal enhancement.[31]

1.4.1. Principle of one-dimensional flow rate modeling for device design

Introduced by Gosselin et al.,[32] theoretical studies were performed to investigate the dynamics of capillary flow using COMSOL Multiphysics as modeling software. Based on the Lucas-Washburn's equation, the flow rate of reagents was simulated according to the capillary pressure in cylindrical and rectangular channels. Particularly, the dynamic of capillary filling with precise time dependency was studied. They concluded that the dimension of the flow channel plays a significant role in capillary flow rates. With a smaller dimension of the channel, a slower filling time was observed. Verified by Byers et al.,[31] thesis findings were incorporated with Darcy equation to describe the flow dynamics as velocity over distance in a variety of membrane dimensions. In other words, it is possible to control the flowing sequence of reagents as their flowing distances and membrane dimensions can be designed accordingly.

1.5. Binding kinetics for LFIA

The effectiveness of the binding interaction in LFIA is a key parameter to evaluating LFIA sensitivity. Surface plasmon resonance (SPR) has been considered as the benchmark technique for deriving real-time kinetic and binding curves (e.g., sensorgram) measurement.[33] Starting from immobilizing antibodies onto the sensor chip surface, the buffer containing target analyte flows over its surface. Their binding interaction would then be evaluated by measuring the change of refractive index over time. As such, the kinetic association (k_a), dissociation (k_d), and equilibrium constants (k_{eq}) of the antibodies can be obtained. With the development of microarray technologies, the field has developed from optical detection methods to microarray-based kinetic assays, which enable the simultaneous measurement of affinity constants over a substantial number of ligands.[34]

1.5.1. Langmuir binding model

Several binding models, such as the Langmuir,[35] heterogeneous analyte and ligand,[36] and bivalent analyte binding models[37], have been developed to describe different biomolecular interactions. Notably, the binding interaction for an antibody and its target analyte is a one-to-one interaction, which indicates that only one ligand interacts with the target analyte at a time. Theoretically, the formation of the antibody-analyte complex can follow pseudo-first-order kinetics since the analyte concentration remains constant after the sample addition. Thus, Langmuir binding model is the simplest model that is suitable for the kinetic analysis of immunoassays. This model divides the whole process into the association phase, equilibrium phase, and dissociation phase, and further describes the rate of complex formation over time with mathematical equations. With a known initial target concentration and kinetic constants of ligand, their relative binding over time can be simulated. The general sensorgram for all three phases is shown in Figure 6.

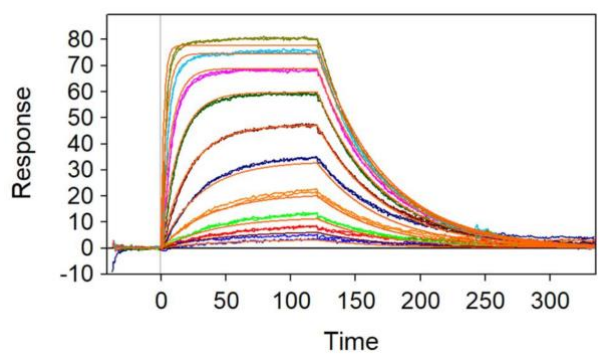

Figure 6. The typical sensorgram phases showing Langmuir binding interaction between ligand and its target in a series of concentrations. Image retrieved from https://www.biofizik.com/surface-plasmon-resonance/.

2. METHODS AND THEORIES

2.1. Computational method for LFIA flow rate analysis

To characterize capillary flow for all components (i.e., three channels) in the LFIA design, one-dimensional flow rate simulations were performed to establish appropriate channels' dimensions for sequential flow control. An extension of Lucas-Washburn and Darcy equations[32] as previously demonstrated by Gosselin et al. was applied. The flow rate of each of the reagents (i.e., nanoparticles, PBS buffer, and amplification reagents) was evaluated in terms of capillary pressure according to the interpretation of Byers' research[31]. Assuming that the cross-section of the nitrocellulose channel is rectangular, capillary pressure (P_c) can be presented by **Eq. 1**:

$$P_c = \frac{\gamma(\cos(\theta_a)) + cos(\theta_b))}{h} + \frac{\gamma(\cos(c)) + cos(\theta_d))}{w} \quad (1)$$

in which γ is the surface tension, θ_i is the angle of fluid in contact with each i surface, h and w are the height and width of the capillary channel.[38] As we assumed a rectangular cross-section of the channel and all reagents are diluted in water, **Eq. 1** can be simplified as:

$$P_c = \gamma\left(\frac{2\cos(\theta)}{h} + \frac{2\cos(\theta)}{w}\right) \quad (2)$$

Here, each surface i is the nitrocellulose channels membrane and θ is the advancing contact angle of pure water on rectangular nitrocellulose. Noted that the nitrocellulose membrane we used has a pore size of 0.45 μm, the advancing contact angle for water on 0.45 μm pore size nitrocellulose was measured to be 40° by Byer et al.[31] While substituting P_c from the **Eq. 2** into the capillary force: $F_{cap} = P_c \times hw$, the equation becomes
$F_{cap} = \gamma 2\cos(\theta)\,(w + h)$. As the nitrocellulose channel with a pore size of 0.45 μm was regarded as small capillary channels, the viscous forces are dominant and inertial effects were neglected, resulting in laminar flow at low Reynolds numbers. Then the capillary force was estimated to be equal to drag force $F_{cap} = F_{drag}$ (i.e., $\gamma 2\cos(\theta)\,(w + h) = 6\mu VLC$) in which μ is the fluidic viscosity, V and L are the travel velocity and travel distance in the capillary channel, and C is the geometric constant given by: $C = \frac{2h}{w} + \frac{w}{h}$. Rearranging the equation to solve for velocity (V):

$$V = k\frac{2\cos(\theta)(h + w)}{6LC} \quad (3)$$

where k is the constant defined by surface tension (γ) divided by viscosity (μ), which can be determined experimentally for each solution, h and w are maintained as the height and width of the nitrocellulose channel. Assuming that each individual channel in a bundle of capillary tubes is evenly distributed in the nitrocellulose membrane, **Eq. 3** can be magnified to the scale of membrane height, width, and length that is correlation to the channel height, width, and distance.[31]

Meanwhile, the velocity to a position can be expressed as the first derivative of position with respect to time $V = dL/dt$ and describe by:

$$\int V(t)dt = \int dL = L + \overline{L} \quad (4)$$

where V and L are the travel velocity and travel distance and t is the required time.

Due to laboratory closure, flow parameters (k values for nanoparticles, chromogenic substrates in a variety of nitrocellulose channels' width (w)) in this simulation were taken from the supporting information experimented by Byers and co-workers.[31] Meanwhile, linear interpolation was incorporated to construct new data points within the range of a discrete set of known values that have not been reported. In the application of this simulation, the new value k can be given from the equation of slope:

$$\frac{k - k_1}{w - w_1} = \frac{k_i - k_1}{w_i - w_1} \quad (5)$$

Here, $k = \gamma/\mu$, and w is the width of the nitrocellulose channel, i represents the width with known discrete values. Taken **Eq. 3, 4, and 5** together, a function of velocity (V) over time (t) can be computed to simulate the one-dimensional flow response to different lengths (L) and widths (w) of membranes with respect to the reagents with a known constant (k) value. To simplify the model, their corresponding flow behavior in any component was assumed to be always the same as flowing in the nitrocellulose membrane as it is the major part of the assay. Referring to the commercial lateral flow test, an appropriate time frame and effective parameters were established in Table S1 (Appendix A).

2.2. Kinetic analysis

The kinetic analysis for the specific binding of monoclonal antibody (or nanobody) with its target was carried out using computational methods. To simplify the model, we attempted to focus on the one-site Langmuir reaction kinetic model. It was assumed that the binding system at the test line is not using biotin-streptavidin interaction but conventional antibody capture. In this case, only nanoparticles conjugated with either antibody or nanobody were lyophilized in the conjugate pad but not in the presence of both antibody and nanobody. In other words, only one ligand molecule interacts with one analyte molecule in the reaction mixture. For the interpretation of one-to-one binding, Langmuir binding model was the optimal model to be adopted to describe their binding interactions over time.[39,40] To apply Langmuir binding model, it was assumed that their binding reactions are equivalent and independent to one another; the concentration of target analyte remains constant upon the sample addition; the reaction rate is not limited by the lateral flow. Generally, the binding interaction and complex formation can be described by the law of mass action:[41]

$$[Ag] + [mAb] \underset{k_d}{\overset{k_a}{\rightleftharpoons}} [Ag - mAb]$$
$$[Ag] + [Nb] \underset{k_d}{\overset{k_a}{\rightleftharpoons}} [Ag - Nb] \quad (6)$$

where Ab or Nb represents the monoclonal antibody or nanobody, Ag represents its specific antigen ligand. The rate of complex formation is denoted by the kinetic association constant k_a and the rate of complex decay is indicated by the kinetic dissociation constant k_d. The ratio of these two rate constants is the equilibrium constant (K_{eq}) or equilibrium dissociation constant ($K_D = 1/k_{eq} = k_d/k_a$) between the target and recognition element.

$$K_{eq} = \frac{k_a}{k_d} = \frac{[Ag - mAb]}{[Ag][mAb]}\ ;$$
$$K_D = \frac{1}{K_{eq}} = \frac{[Ag][mAb]}{[Ag - mAb]}$$

or (7)

$$K_{eq} = \frac{k_a}{k_d} = \frac{[Ag - Nb]}{[Ag][Nb]}\ ;$$
$$K_D = \frac{1}{K_{eq}} = \frac{[Ag][Nb]}{[Ag - Nb]}$$

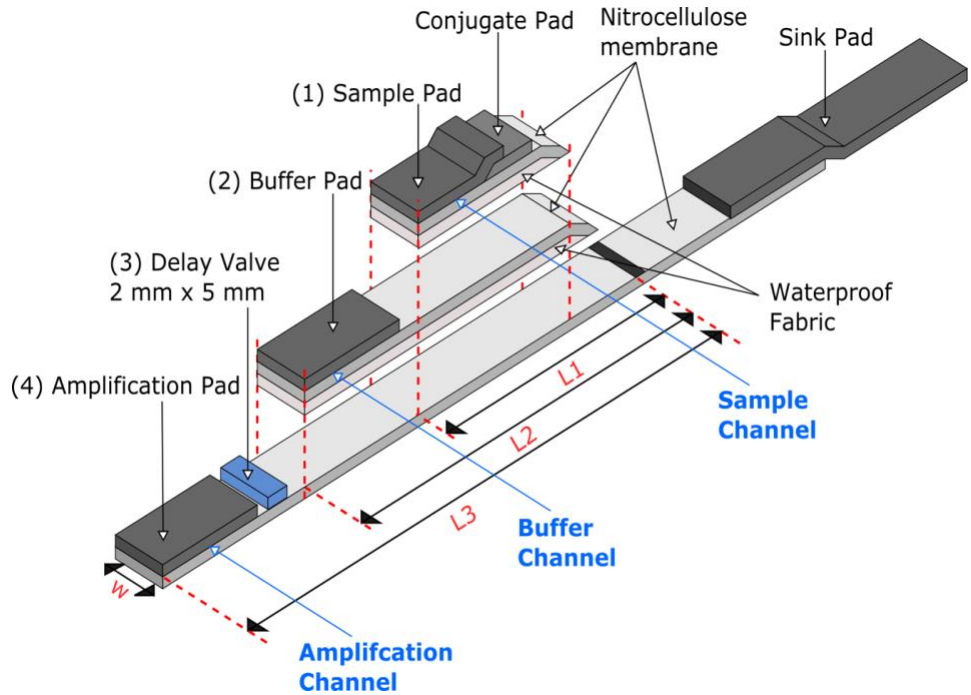

Figure 7. Proof-of-concept of a dimensionally undefined LFIA device in three channels design with amplification, buffer, and sample channels (from bottom to top), where $L1, L2, L3$ represent the length from the far end of the respective channel to the test line, w indicates the width of the assay

2.2.1. Association phase in LFIA

Two main events are associated with the kinetic association phase in the LFIA design. The binding interaction between a specific antibody or nanobody with its target analyte occurs as the mixture flows along the membrane. When the mixture reached the test line, another binding interaction is initiated by another antibody or nanobody, which is specific for another epitope of the analyte and is bio-printed on the test line to capture the former complex. During these two events, the fraction of binding (θ) over time (t) can be derived to **Eq. 8** by taking derivative of the above **Eq. 6 & 7** with respect to time.[34]

$$\theta = \frac{R_t}{R_{max}} = \frac{[A]}{K_D + [A]}(1 - e^{-(k_a[A]+k_d)t}) \quad \textbf{(8)}$$

Here, θ is the fraction of binding, R_t is the signal level at a certain time (t) , R_{max} is the maximum signal level corresponding to the maximum concentration of antibodies or nanobodies (i.e., $R_{max} \propto [MAb]_{max}$ or $R_{max} \propto [Nb]_{max}$), k_a and k_d is the kinetic association constant and kinetic dissociation constant, K_D is the equilibrium dissociation constant given by $K_D = 1/k_{eq} = k_d/k_a$, and $[A]$ is the initial concentration of target analyte. As shown in the **Eq. 8**, the fraction of binding (i.e., the ratio of complex formation) over time is proportional to K_D which determines the equilibrium level at the former term ($[A]/(K_D + [A])$). The latter term ($1 - e^{-(k_a[A]+k_d)t}$) describes the complex formation in terms of response units which indicates the level of response at equilibrium and the time required for the reaction to reach equilibrium. It is worthy to note that the testing time is generally too short for the binding reaction to reach equilibrium in LFIA.

2.2.2. Dissociation phase in LFIA

In our established timeframe, the PBS buffer solution is introduced at the same time as the sample solution onto the LFIA device at t = 0. When the buffer solution reaches the test line at t = 300 s, the unbound target analytes are removed from the flow and drops to zero concentration. This sudden removal of targets in the reaction mixture results in the start of kinetic dissociation. The rate of complex dissociation follows exponential decay kinetics as shown below.[34]

$$R_t = R_0 e^{-k_d t} \quad \textbf{(9)}$$

R_t is the signal level at time (t), R_0 is the signal level at the end of the association phase, and k_d is the kinetic dissociation constant of the ligand. Similar to **Eq.** 8, this equation describes the time required to reach a response level during the dissociation phase. Likewise, the equation can be expressed by the fraction of dissociation θ_t following the same response level.

$$\theta_t = \theta_0 e^{-k_d t} \quad \textbf{(10)}$$

where the θ_0 the fraction of binding at the end of the association phase.

3. RESULTS AND DISCUSSION

3.1. LIFA device design

The LFIA device was designed according to the laboratory assembly practice of lateral flow test strips (Figure 7). The device comprised of three channels that controlled the reagents' flow in three corresponding steps, including “target binding” in the sample channel (Figure 8A), “washing” in the buffer channel (Figure 8B), and “signal amplification” in the amplification channel (Figure 8C). As such, components of the LFIA strip is assembled by lamination from the bottom to the top. Notably, each channel is assembled with waterproof plastic in between to prevent flow contamination. A 5 µL “delay valve” was designed at 1 mm away from the amplification pad. Due to the saturation of the amplification channel, capillary pressure

exerted in the junction would prevent backflow from sample and buffer channels during the lateral flow test.[42]

An appropriate timeframe was created to standardize the nanocatalyst-amplified LFIA test (Table S1). With known flow parameters of reagents, the use of Washburn flow modeling could indicate the required dimensions for each of the channels. A sequential delivery of reagents within the respective time frame can thus be achieved. Due to the laboratory closure, this measurement focused on the flow parameters (i.e., surface tension and viscosity) that are available in the literature. The flow modeling was simulated by using antibody-conjugated AuNPs as the recognition element, and DAB as the chromogenic substrate. Meanwhile, their flow parameters were integrated with linear interpolation (**Eq. 5)** to estimate corresponding k values for AuNP and DAB solutions in 1–5 mm nitrocellulose channels (Table S2).

3.1.1. Device dimensions

The dimensions of the LFIA device was defined based on the simulation results of the one-dimensional flow rate modeling. To measure the one-dimensional flow rate, it is necessary to clarify the flowing sequence of reagents with respect to time. Firstly, the flow rate measurement was simulated for the sample channel (Figurer 8A), of which the solution mixture contains lyophilized antibody-conjugated nanoparticles, orthogonally biotinylate nanobody-biotin, and target analyte. As the nanoparticle is the only substrate (which acts as surfactants) that would affect the surface tension and viscosity, the flow parameters of other molecules could be neglected. According to the **Eq. 3 & 4**, the flow rates of the dominant surfactant (i.e., AuNP) in 1–5 mm wide sample channels were simulated as a function of velocity over distance and distance over time (Figure 9) based on the k values from Table S2. For the one-dimensional flow rate simulation of the buffer channel (Figure 8B) and amplification channel (Figure 8C), PBS buffer and DAB solution contained neither surfactants nor nanoparticles. They were assumed to have the same surface tension and viscosity as the DAB solution which is generally regarded as the same for pure water. Their Washburn flow behavior were simulated as a function of velocity over distance and distance over time (Figure. 10) by referring to the k' values indicated in Table S2.

Incorporating the simulation results with the established time frame, the required distances for each channel to achieve sequential flow control is shown in Table 1. Assuming the liquid does not bypass the structure of membrane and overflow, the width of all channels was set to be 4 mm since it produced a relatively fast and consistent flow (Figure 9 & 10). Compared with the 5 mm wide membrane, the corresponding lengths for the 4 mm membrane are more practical for fabrication in general manufacturing practice of lateral flow strip (e.g., commercialized pregnancy test strip dimension [110.7 x 7.6 x 58.4 mm; ClinicalGuard®]). Its total size is also portable for PoC settings. As a result, the length from the far end of sample, buffer, and amplification channels to the test line (i.e., $L1, L2, L3$) were defined as 35, 64, and 90 mm (Figure 11). Theoretically, the one-dimensional flow in each of the solutions will reach the test line at the specific time if the membrane dimension is fabricated accordingly. Particularly, a 5 µL "delay valve" (2 X 4 mm^2) was designed to prevent backflow of the sample to the amplification channel, which might result in the catalyzed reaction outside the detection zone.

Table 1. Simulation results demonstrating the device dimension correlated to each reagent flowing in sample, buffer, and amplification channel.

Channels	Sample	Buffer	Amplification
Main types of solution	AuNP	PBS	DAB
Time (s)	120	300	600
Length / Width	***L*1 (mm)**	***L*2 (mm)**	***L*3 (mm)**
5 mm	37.43	70.89	100.46
4 mm	34.16	63.11	89.46
3 mm	30.56	52.23	76.90
2 mm	27.92	46.36	65.77
1mm	21.79	29.03	41.26

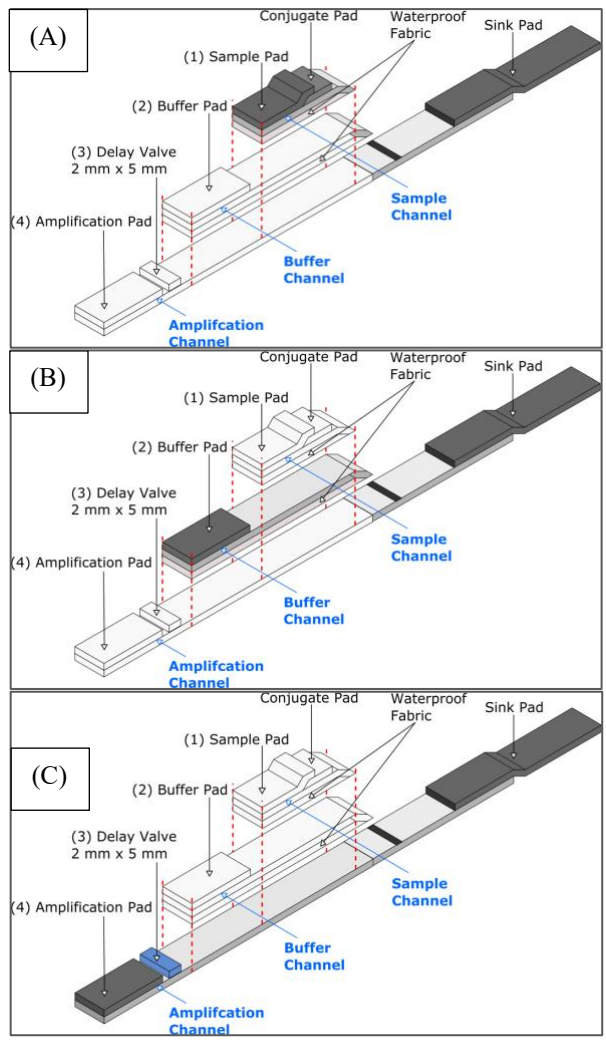

Figure 8. Schematic illustration of the LFIA device showing three different channels for respective flow of reagents. (A) Sample channel involving specific binding of target; (B) buffer channel involving PBS buffer for washing; (C) amplification channel involving the catalyzation of colorimetric reaction of chromogenic substrates in the presence of H_2O_2.

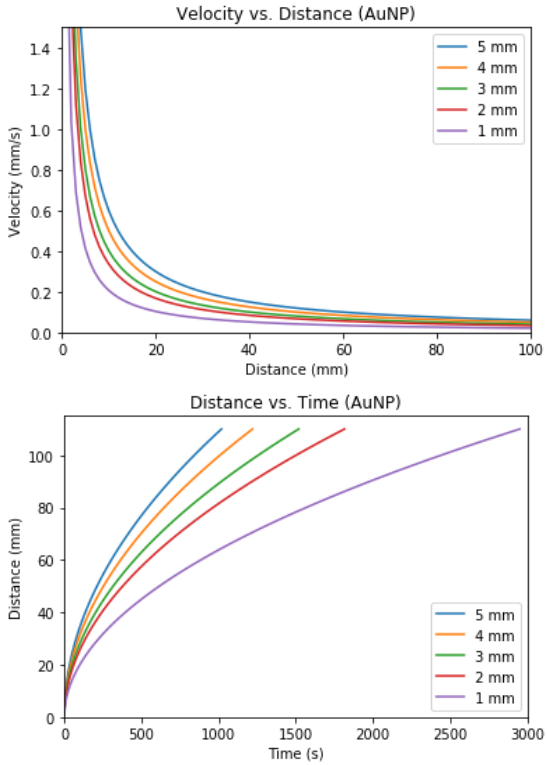

Figure 9. Simulation result of AuNP solution as velocity over distance (top) and distance over time (bottom) in 1–5 mm wide channels.

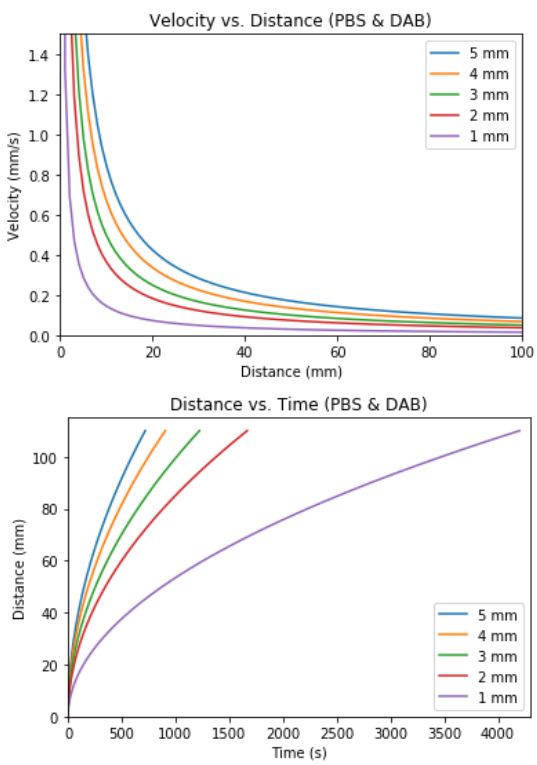

Figure 10. Simulation result of PBS and DAB solutions as velocity over distance (top) and distance over time (bottom) in 1–5 mm wide channels.

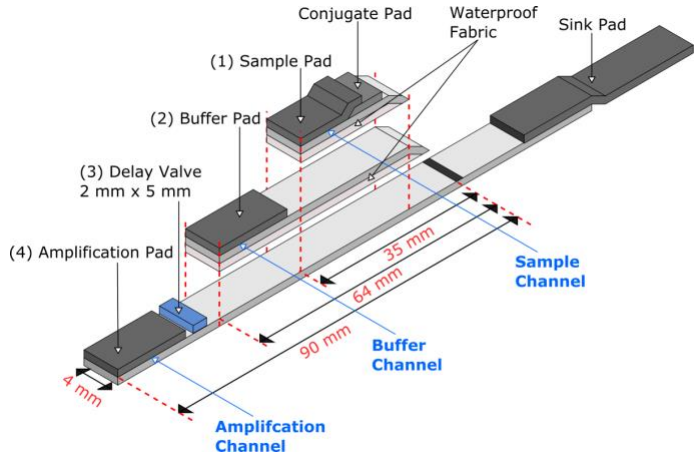

Figure 11. Schematic showing the dimensionally defined LFIA device composed of three channels design with amplification, buffer, and sample channels (from bottom to top).

3.2. Binding kinetics

To study the binding kinetics in the LFIA device, the binding process was simulated to describe the binding interaction between monoclonal antibodies as well as nanobodies and their target p24 capsid protein. As the LFIA test could never reach chemical equilibrium, recognition or capture elements with a relatively higher affinity than the ones used in conventional immunoassay (e.g. ELISA) is preferred to ensure the assay efficacy. According to the established timeframe (Table S1), a series of target concentrations (i.e., [p24]) in different orders of magnitude were used to simulate the binding interaction of NIH-3537, CP-HIV-1/2, 59H10, and 37E7 towards based on their kinetic parameters reported by Gray et al. (Table S3).[43]

The order of magnitude for [p24] that can achieve a 90% fraction of binding was investigated based on the designed time frame. For the first association binding, the antibodies and nanobodies interacts with the target for 120 seconds before the solution mixture reaches the test line. It was assumed that their binding reaction is initiated upon the sample addition; there are only one-site Langmuirian binding interactions between them and the target concentration remains constant; their reaction rates are not limited by mass transport. The kinetic simulation results for NIH-3537 (Figure 12A), CP-HIV-1/2 (Figure 12B), 59H10 (Figure 12C), and 37E7 (Figure12D) were plotted as a function of fraction of binding over time. As shown in Figure 12, their kinetic bindings in response to time are highly dependent on their kinetic constants. Nanobody 59H10 demonstrates the relative highest affinity towards p24 as it has the highest affinity (i.e., slopes) in all target concentration levels. This result also verifies the fact that 59H10 nanobody has the relative lowest kinetic equilibrium dissociation constant (K_D) compared to others. The minimum orders of magnitude for NIH-3537, CP-HIV-1/2, 59H10, and 37E7 to achieve at least 90% of binding in 120 s are shown as 10 μM, 1μM, 100 nM, and 1μM.

However, the clinically relevant ranges of [p24] are far below the 1 μM for the early HIV detection, especially in acute-phase HIV plasma samples.[44] Theoretically, it is impossible to achieve a high level of binding between molecules and its cognate ligand while in the application of LFIA, which has a short timeframe for full interaction. Developing novel

biomolecules that display picomolar to femtomolar range s would be

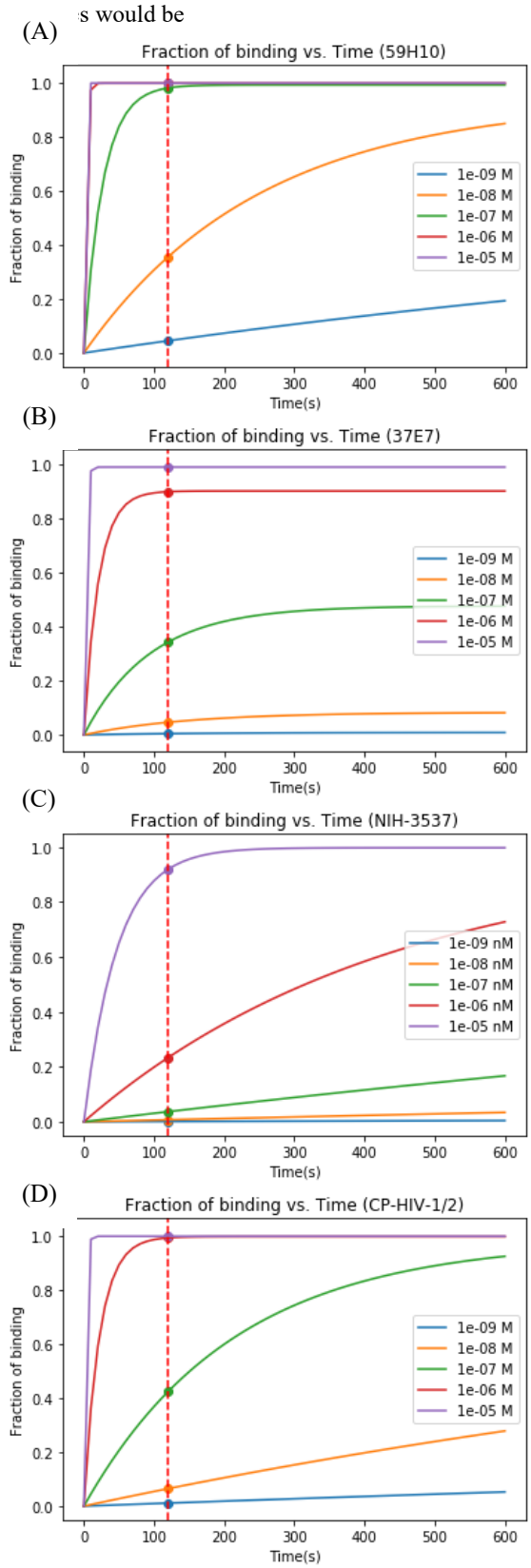

Figure 12. Kinetic simulation of nanobodies (A) 59H10, (B) 37E7, and antibodies (C) NIH-3537, (D) CP-HIV-1/2 towards a series of concentration of p24 (1 nM, 10 nM, 100 nM, 1 μM, 10 μM) as a function of fraction of binding (θ) over time (t). The vertical dash line (red) indicates the time at t = 120 s when t = 0 is defined as the time of sample addition. The intersections represent their corresponding fraction of binding at that time.

a scientific breakthrough in improving the sensitivity of LFIA. In spite of that, we further evaluated binding interaction between 59H10 and p24 in a new series of [p24] that are within clinically relevant ranges. Its fraction of binding in response to time was significantly increased whenever the target concentration was doubled (Figure 13). This implies that increasing the concentration of target analytes from patient samples by additional approaches, such as incorporating with host antibody detection, would be an alternative to increasing the sensitivity of LFIA tests.

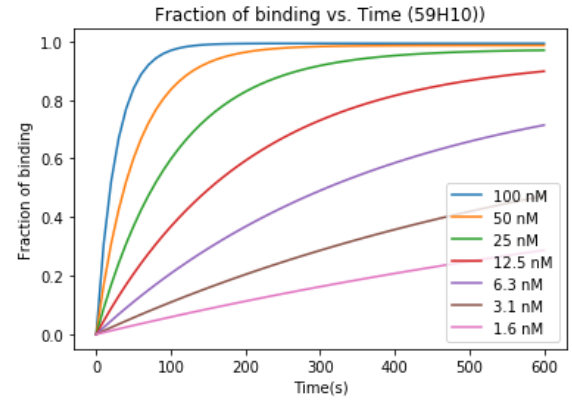

Figure 13. Simulation results showing effect of target concentrations on the fraction of binding over time for 59H10 nanobody.

The goal of this thesis is to exploit computational model for the kinetics studies of biomolecular recognition element and its specific target with known kinetic parameters. At the start, it was assumed that the binding interaction was equivalent and independent for all binding sites and the complex formation complied with first-order kinetics. In other words, the target analyte might have more than one epitope that can bind to the same antibody/nanobody, but only one-to-one binding can occur at each process. In this case, the binding association kinetics (**Eq. 8**) and dissociation kinetics (**Eq. 10**) for previous antibodies and nanobodies were simulated in application to the developed LFIA design.

To determine their corresponding binding level at the end of each flow sequence, the whole detection process was separated into two phases at an initial target concentration of 12.5 nM. During phase 1, the sample containing target was applied onto the (1) sample pad and the buffer solution was applied onto (2) buffer pad, (3) delay valve, and (4) amplification pad at t_0= 0 s (Figure 11). The binding interaction between a recognition element and p24 proceeded up till the sample solution mixture reached the test line at t = 120 s. During phase 2a, the target-antibody/-nanobody complex from phase 1 interacted with the bio-printed antibodies/nanobodies on the test line for 180 s until the buffer solution arrived at the test line at t = 300 s. Herein, the initial target concentration became $[Ag_2] = [Ag_1]\theta_1$, in which $[Ag_1]$ is the initial target concentration of 12.5 nM, $[Ag_2]$ is the complex concentration at the beginning of phase 2a, and θ_1 is the fraction of binding at the end of phase 1. During phase 2b, the buffer solution reached to the test line, resulting in a rapid drop of complex concentration until the arrival of amplification reagents at t = 600 s. Its fraction of binding then became $\theta_3 = \theta_2 e^{-k_d t}$, where θ_2 is the fraction of

binding at the end of phase 2a. Hence, the final concentration of complex deposited at the test line can be calculated by $[Ag_1]\theta_1\theta_3$.

Table 2. Quantitative results of the kinetic studies of 59H10, 37E7, NIH-3537, and CP-HIV-1/2 in the case of the developed LFIA device (computed from Jupiter Notebook).

Phases	Binding ligands	Initial target concentration $[Ag]$	Fraction of binding (θ) in t (s)	Final complex concentration $[Ag']$
Phase 1 association (t = 120 s)		$[Ag_1]$	θ_1	$[Ag_1'] = [Ag_1]\,\theta_1$
	59H10	12.5 nM	4.22 x 10^{-1}	5.28 nM
	37E7	12.5 nM	5.76 x 10^{-2}	7.20 x 10^{-1} nM
	NIH3537	12.5 nM	8.43 x 10^{-3}	1.05 x 10^{-1} nM
	CPHIV1/2	12.5 nM	7.75 x 10^{-2}	9.69 x 10^{-1} nM
Phase 2a association (t = 180 s)		$[Ag_2] = [Ag_1']$	θ_2	$[Ag_2'] = [Ag_2]\theta_2$
	59H10	5.28 nM	2.94 x 10^{-1}	1.55 nM
	37E7	7.20 x 10^{-1} nM	4.46 x 10^{-3}	3.21 x 10^{-3} nM
	NIH3537	1.05 x 10^{-1} nM	1.44 x 10^{-4}	1.52 x 10^{-5} nM
	CPHIV1/2	9.69 x 10^{-1} nM	1.62 x 10^{-2}	1.57 x 10^{-2} nM
Phase 2b dissociation (t = 300 s)			$\theta_3 = \theta_2 e^{-k_d t}$	$[Ag_3'] = [Ag_2]\,\theta_3$
	59H10		2.69 x 10^{-1}	1.42 nM
	37E7		6.54 x 10^{-4}	6.54 x 10^{-4} nM
	NIH3537		1.37 x 10^{-4}	1.44 x 10^{-5} nM
	CPHIV1/2		1.52 x 10^{-2}	1.48 x 10^{-2} nM

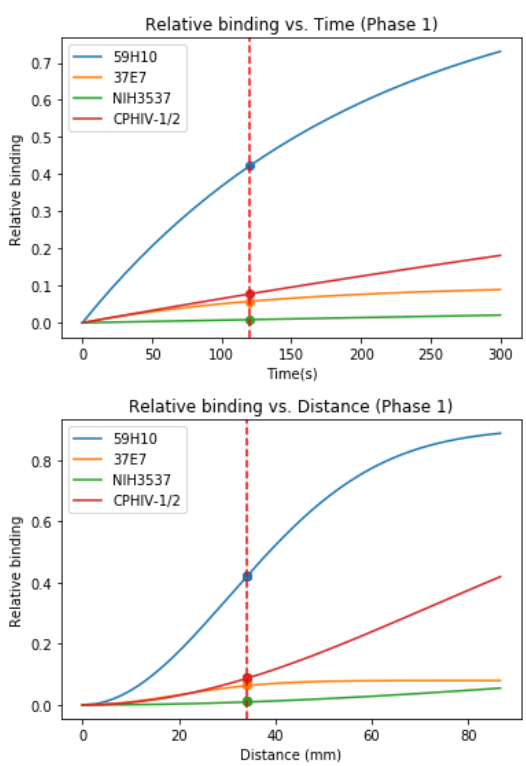

Figure 14. Simulation result of the kinetic analysis over time (left) and kinetic analysis over distance in phase 1, only involving the binding between targets and nanoparticles-conjugated 59H10, 37EC, NIH-3537, or CP-HIV-1/2. Their corresponding binding levels intersect with the vertical dotted line (red) at t = 120 s (left) and at L_1= 34.1 mm (right). Their kinetic analyses over distance (right) were obtained by incorporating with Washburn flow measurement of AuNP to evaluate their binding performance in a variety of lengths from the sample pad to the test line.

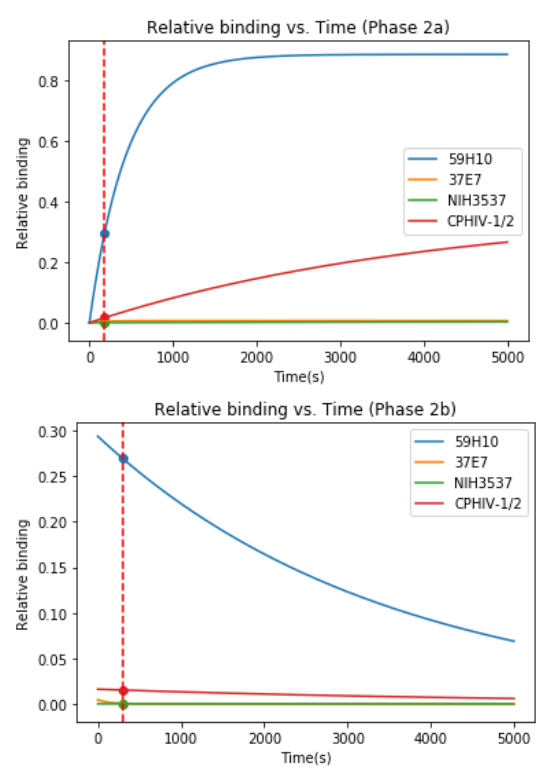

Figure 15. Simulation result of the kinetic analysis in phase 2a (top), considering the binding interaction between the complex formed in phase 1 and immobilized 59H10, 37EC, NIH-3537, or CP-HIV-1/2 on the test line. Simulation result in phase 2b (bottom), where the new sandwich complex formed in phase 2 dissociated due to the arrival of the buffer solution. Their corresponding binding levels intersect with the vertical dash line at $t = 180$ s (top) and $t = 300$ s (bottom). Phase 2 and phase 3 is an ongoing process related to the association and dissociation of the complex.

The simulation result (Figure 14 & 15, Table 2) demonstrates the promising applicability of 59H10 nanobody and CP-HIV-1/2 antibody in absorbance-based LFIAs. Their relative higher affinity binding ensures a considerable amount of complex being captured by the test line. In application to the detection of p24, this kinetic simulation provides numerical evidence that antibody CP-HIV-1/2 is more favorable than NIH3537 as the conjugate being functionalized with nanocatalysts; integration with biotinylated nanobody 59H10 is more ideal for facilitating nanocatalyst-analyte complex binding to the streptavidin test line. Particularly, this simulation strategy may be adapted to analyze other protein-ligand for the detection of other molecules, such as bacterial protein and nucleocapsid protein. Another significance of this basic model is to computationally estimate the final concentration of nanocatalyst-target complex deposited at the test line, which is highly related to the efficiency of signal amplification in LFIA.

4. CONCLUSION

By utilizing Washburn flow and Langmuir binding simulations, a theoretical model was established. Within the REASSURE criteria, this device demonstrated the feasibility of achieving a fully automated nanocatalyst-amplified LFIA for PoC application. This thesis validated that the synthesized PtNPs exhibited intrinsic peroxidase-like activity that catalyzed the oxidation of chromogenic substrates in the presence of H_2O_2. The effectiveness of their catalytic activity was also dependent on difference types of chromogenic substrates, including TMB, OPD, and ABTS. In optimal pH conditions, PtNPs demonstrated a stronger catalytic ability toward TMB than OPD and ABTS (Appendix B). The development of new chromogenic substrates, which are effective to the employed nanocatalyst, is one of the alternatives to enhance the sensitivity of LFIA biosensing.

For the device design, the Washburn flow theory and mathematical model were utilized to simulate the flow of LFIA components according to the designed timeframe. A three-channel design (sample channel, buffer channel, and amplification channel) was established to standardize the testing procedures of colorimetric LFIA into three steps, including target processing, washing, and signal amplification. The device dimensions were defined based on the Washburn flow simulation results in different widths of the reaction membrane. The experiment simulated the flow of respective reagents as a function of their flowing velocity (V) over travel distance (L) in the capillary channel and computed the function into travel distance (L) over time (t). It was concluded that a 4 mm wide device with a 35 mm sample channel, 64 mm buffer channel, and a 90 mm amplification channel could theoretically achieve full automation of sequential flow at precisely timed intervals. Additionally, this report studied the binding kinetics of an ideal single-domain binding model. Simulations of the binding response of nanobodies (59H10, 37E7) and monoclonal antibodies (NIH-3537, CP-HIV-1/2) toward an ideal target over time showed that the initial target concentration and recognition elements' affinity would significantly affect the final concentration of signal amplification labels (i.e., nanocatalyst deposited at the test line), which is the key to the amplifying efficiency of LFIA test.

Nevertheless, several improvements can be made in the device design. Whenever laboratory access is possible, the volume capacity of the sample pad, nitrocellulose membrane, and conjugate pad should be considered by calculating the difference between dry mass and wet mass of the membrane. Clean delivery of the developed model should be examined by using coloring techniques to demonstrate the sequential flow control and validate its proof-of-concept. For the kinetic studies, the simulation of the bivalent analyte binding model (in which one analyte molecule can bind to one or two ligand molecules) is more appropriate in the application of the kinetic analysis of LFIA that deployed biotin-streptavidin capture.

ACKNOWLEDGMENT

I would like to express my gratitude to my supervisor, Professor Molly Stevens for her advice, guidance, and encouragement throughout the MRes program. I was able to adapt, acquired valuable experience, and overcome difficulties in this incredibly challenging time. Her dedication to pursuing the excellence in medical research has inspired me tremendously.

I am also grateful for my postdoctoral and doctoral mentors, Dr. John Goertz and Brian Chan for their thoughtful mentorship, and most importantly their care and empathy for my challenging situation this year. I would especially like to thank Brian Chan for his guidance in the aspect of laboratory-based work, where I have honed my professionalism in preparing and executing my experiments. I have learned how to conduct nanomaterials synthesis, modification, and application. Meanwhile, Dr. John Goertz guided me through the computational aspect of my project, which became the major part due to the college's closure. He demonstrated the significance of thermotical studies in academic research and gave me valuable insight to help with my computational work. I would not be able to finish my MRes project without both of their support.

Lastly, my biggest thanks go to the Department of Bioengineering of Imperial College London, which accepted my MRes application; provided me access to the college and laboratories as well as the resources to scientific seminars, workshops, and events that significantly helped me to grow into a better individual.

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Appendix A: Supporting information

Table S1. Established timeframe of the designed LFIA device showing its dimensional parameters that determine the sequential flow control. $L1, L2, L3$ and w refer to the length and width highlighted in Figure 7.

Time (s)	Events	Dimensional parameters	Channel
t = 0	Addition of sample and buffer		
t = 120	Complexation of sample and Ab-PtNC conjugates reaches the test line	$L1, w$	Sample channel
t = 300	Buffer solution reaches the test line	$L2, w$	Buffer channel
t = 600	Amplification solution (CN/DAB, H_2O_2) reaches the test line	$L3, w$	Amplification channel

Table S2. Characterized flow parameters of gold nanoparticle (AuNP) and 3,3'-diaminobenzidine (DAB) solutions. Capitalized from Byers et al.,[31] table was integrated with linear interpolation to estimate the values in 3 and 4 mm wide nitrocellulose channels.

Characterized Parameters	**Value**
40 nm AuNP in 5 mm wide nitrocellulose channel	$k_5 = 230.3$
40 nm AuNP in 4 mm wide nitrocellulose channel	$k_4 = 191.5$
40 nm AuNP in 3 mm wide nitrocellulose channel	$k_3 = 152.7$
40 nm AuNP in 2 mm wide nitrocellulose channel	$k_2 = 126.1$
40 nm AuNP in 1 mm wide nitrocellulose channel	$k_1 = 75.16$
DAB in 5 mm wide nitrocellulose channel	$k'_5 = 326.4$
DAB in 4 mm wide nitrocellulose channel	$k'_4 = 258.0$
DAB in 3 mm wide nitrocellulose channel	$k'_3 = 189.6$
DAB in 2 mm wide nitrocellulose channel	$k'_2 = 137.2$
DAB in 1 mm wide nitrocellulose channel	$k'_1 = 52.76$

Table S3. Summary of the kinetic parameters of two nanobodies (Nbs) and monoclonal antibodies (mAbs). Table adapted from Gray et al.[43]

	mAbs		Nbs	
Kinetic parameters	NIH-3537	CP-HIV-1/2	59H10	37E7
K_D (M)	2.2×10^{-8}	1.6×10^{-9}	6.9×10^{-10}	1.1×10^{-7}
k_a ($M^{-1}s^{-1}$)	2.1×10^{3}	4.5×10^{4}	3.7×10^{5}	4.2×10^{4}
k_d (s^{-1})	1.7×10^{-4}	2.0×10^{-4}	2.9×10^{-4}	6.4×10^{-3}